



Modelling in Mechanics · Worksheet 1

Mechanics Chapter 1: Modelling in Mechanics. Answer all questions. Show your working clearly. Total: 40 marks. Time: 50 minutes.

Name _____

Class _____

Date _____

INSTRUCTIONS

- Answer **all** questions in the spaces provided. Show all your working.
- There are two sections, one for each topic in Chapter 1. Within each section the questions increase in difficulty; the **last question** of each section is set at A level exam standard.
- Take $g = 9.8 \text{ ms}^{-2}$. When you comment on a model, name the **assumption** and explain its effect **in context**.
- The **Roar-level challenge** at the end is a harder stretch question.
- The marks for each question are shown in brackets, e.g. [3].

Section	Marks available	Marks scored
A Mathematical models and modelling assumptions	17	
B Quantities, units, and scalars vs vectors	19	
Roar-level challenge	4	
Total	40	

Section A Mathematical models and modelling assumptions

17 marks

1. A string is described as *light* and *inextensible*. Explain what each of these words means. [2]

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2. A tennis ball is hit across a court and is modelled as a particle. State one thing that this modelling assumption ignores. [2]

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3. A book is pushed across a table. In a model, the book is treated as a particle and the table as smooth. [3]

(a) Explain what is meant by the table being *smooth*.

(b) Explain why the table being smooth is unlikely to be a good assumption, and how this affects the book's motion.

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4. A crate is pulled across a floor by a rope. The rope is modelled as light and inextensible, and the crate is modelled as a particle. [3]

Explain how each of these three assumptions simplifies the problem.

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5. A stone is dropped from a bridge into a river. A student models the motion by treating the stone as a particle falling freely under gravity with $g = 9.8 \text{ m s}^{-2}$.

(a) State two modelling assumptions the student has made.

[2]

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(b) Explain how including air resistance would change the time taken for the stone to reach the water.

[1]

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(c) Suggest two ways the model could be made more realistic.

[2]

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(d) Explain why the same model would not be suitable for a leaf falling from the bridge.

[2]

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(Total for Question 5 is 7 marks)

**Section B Quantities, units, and scalars vs vectors**

19 marks

6. State whether each quantity is a scalar or a vector: velocity, time, force, distance. [2]

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7. Convert (a) 72 km h^{-1} into m s^{-1} , (b) 450 g into kg, (c) 2.5 minutes into seconds. [3]

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8. (a) Find the weight of a particle of mass 5 kg. [3]

(b) A box has weight 147 N. Find its mass.

(c) On the Moon, $g = 1.6 \text{ ms}^{-2}$. Find the weight of the 5 kg particle on the Moon.

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9. A walker walks 3 km due east and then 4 km due north. The whole walk takes 1 hour 15 minutes. [4]

(a) Find the total distance travelled and the magnitude of the displacement.

(b) Find the average speed and the magnitude of the average velocity, in km h^{-1} .

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10. A ball is thrown vertically upwards. It rises 12 m and then falls back to the point it was thrown from. The whole motion takes 3 s.

(a) Find the total distance travelled by the ball and its displacement at the end of the motion. [2]

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(b) Find the average speed and the average velocity of the ball over the 3 s. [2]

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(c) A car of mass 1200 kg travels at 90 km h^{-1} . Find its speed in m s^{-1} and its weight. [2]

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(d) Explain why the car's weight is a vector but its mass is a scalar. [1]

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(Total for Question 10 is 7 marks)



ROAR-LEVEL CHALLENGE

4 marks

11. A runner runs once round a circular track of radius 50 m in 62.8 s, starting and finishing at the same point. [4]
- (a) Find her average speed and her average velocity for the whole lap.
- (b) Find the magnitude of her displacement and of her average velocity after half a lap, assuming she runs at constant speed.

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END OF WORKSHEET

Total: 40 marks